

# Probability - Homework 2

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**2.1-7**

(a)

(b)

(c)

(d)

## 2.2-10

**2.2-11**

## 2.3-16

(a)

Let the pmf be  $f(x)$ . If  $X = 0, 1$ ,  $P(X = 1) = f(1) = 0$ ,  $P(X = 0) = f(0) = 0$ . For  $X > 1$ , we can observe that there are only two possibilities, one is the last two flip being both head, the other one is the last two flip being both tail. The two possibilities yield exactly two distinct permutations (e.g. for  $X = 4$ , the only results are  $(h, t, h, h)$  and  $(t, h, t, t)$ , where the tuple  $(\dots)$  represents the result of each round,  $h$  means head, while  $t$  means tail).

Hence, when  $X = x$  ( $x \neq 1$ ), we have  $f(x) = 2 \cdot (\frac{1}{2})^x = 2^{1-x}$  (The probability of one permutation of tails and heads in  $x$  flips is  $(\frac{1}{2})^x$ )

So the pmf  $f$  is:

$$f(x) = \begin{cases} 0 & \text{if } x = 0, 1 \\ 2^{1-x} & \text{if } x \neq 0, 1 \end{cases} \quad (1)$$

(b)

Suppose the mgf to be  $M_X(t)$ , then:

$$\begin{aligned} M_X(t) &= E[e^{tX}] \\ &= \sum_{x=2}^{\infty} e^{xt} f(x) \\ &= \sum_{x=2}^{\infty} e^{xt} 2^{1-x} \\ &= 2 \sum_{x=2}^{\infty} \left(\frac{e^t}{2}\right)^x \end{aligned}$$

The infinite geometric series converge if and only if  $|\frac{e^t}{2}| = \frac{e^t}{2} < 1$ , i.e.  $t < \ln 2$ .

And the sum (if exists) (First term is  $\frac{e^{2t}}{4}$ , common ratio  $r = \frac{e^t}{2}$ ) is obtained by:

$$M_X(t) = 2 \frac{\frac{e^{2t}}{4}}{1 - \frac{e^t}{2}} = \frac{e^{2t}}{2 - e^t} \quad (\text{where } t < \ln 2)$$

(c)

By taking 1st and 2nd derivative, we obtained:

$$\begin{aligned} M'_x(t) &= -\frac{(e^t - 4)e^{2t}}{(e^t - 2)^2} \\ M''_x(t) &= -\frac{e^{2t}(e^{2t} - 6e^t + 16)}{(e^t - 2)^3} \end{aligned}$$

Let the mean be  $\mu$  and the variance be  $\sigma^2$ , then:

$$\begin{aligned}\mu &= M'_X(0) = 3 \\ \sigma^2 &= M''_X(0) - (M'_X(0))^2 = 11 - 3^2 = 2\end{aligned}$$

**(d)**

We can use the results in (a) directly:

$$\begin{aligned}P(X \leq 3) &= P(X = 2) + P(X = 3) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4} \\ P(X \geq 5) &= 1 - P(X \leq 4) = 1 - P(X = 2) - P(X = 3) - P(X = 4) = \frac{1}{8} \\ P(X = 3) &= 2^{1-3} = \frac{1}{4}\end{aligned}$$

**2.4-18**

Let  $X$  be the number of required test,  $n$  be the number of individuals,  $p$  be the probability of a person is having the disease.

The test result of the common pool is negative if and only if all individuals have no this kind of disease, and we only need  $X = 1$  test, also:

$$P(X = 1) = (1 - p)^n$$

The test result of the common pool is positive if and only if at least one individuals have this kind of disease, and we need  $X = n - 1$  test, also:

$$P(X = 1) = 1 - (1 - p)^n$$

Hence, the expected value of  $X$  (given  $p = 0.05, n = 5$ ) is:

$$E[X] = 1 \cdot (1 - p)^n + (n - 1) \cdot (1 - (1 - p)^n) \approx 2.13110$$

The expected value of the number of tests needed is (approximately) 2.13110.

**2.6-9****(a)**

Let  $X$  be the number of request per day, and we are given that  $X$  follows a Poisson distribution with a mean of 3.

Since the store only stocks  $n = 4$  newspaper. Let  $Y$  be the number of newspaper actually sold, then  $Y = \min(X, 4)$ .

We then can calculate the expected value of  $Y$ :

$$\begin{aligned}
 E[Y] &= \sum_{x=0}^3 x \cdot P(X = x) + \sum_{x=4}^{\infty} 4 \cdot P(X = x) \\
 &= 0 \cdot P(X = 0) + P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot P(X \geq 4) \\
 &= 0 \cdot P(X = 0) + P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot \left(1 - \sum_{x=0}^3 P(X = x)\right) \\
 &= 4 - 4 \cdot P(X = 0) - 3 \cdot P(X = 1) - 2 \cdot P(X = 2) - P(X = 3) \\
 &= 4 - 4 \cdot 0.04979 - 3 \cdot 0.14936 - 2 \cdot 0.22404 - 0.22404 \\
 &= 2.68064
 \end{aligned}$$

So the expected of the number sold is 2.68064.

**(b)**

Define the random valuable  $X$  be the number of requests for newspaper per day, and  $f$  be the p.m.f. By the formula for Poisson distribution:

$$P(X \leq n) = \sum_{x=0}^n f(x) = \sum_{x=0}^n \frac{\lambda^x e^{-\lambda}}{x!}, \quad P(X > n) = 1 - P(x \leq n)$$

By some calculation:

$$P(X = 0) = e^{-3} \approx 0.04979$$

$$P(X = 1) = 3e^{-3} \approx 0.14936$$

$$P(x = 2) = \frac{3^2 e^{-3}}{2!} \approx 0.22404$$

$$P(x = 3) = \frac{3^3 e^{-3}}{3!} \approx 0.22404$$

$$P(x = 4) = \frac{3^4 e^{-3}}{4!} \approx 0.16803$$

$$P(x = 5) = \frac{3^5 e^{-3}}{5!} \approx 0.10081$$

$$P(x = 6) = \frac{3^6 e^{-3}}{6!} \approx 0.05041$$

We found:

$$P(X > 5) = 0.08393 > 0.05, P(X > 6) = 0.03353 < 0.05$$

Thus, the minimum number the manager should order is 6.